Plastics, composites & elastomers

D&M WITH MR KELLY

Higher · Materials & manufacture · Worksheet

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ explain the difference between thermoplastics and thermosetting plastics
- ☐ describe the properties and uses of ABS, acrylic, polypropylene and polystyrene
- ☐ describe the properties and uses of urea formaldehyde and melamine formaldehyde
- ☐ describe nylon, polythene and PVC
- ☐ describe elastomers and composites (GRP, carbon fibre)
- ☐ describe methods to identify materials, including what each result tells you
- ☐ explain how materials influence design and manufacture
- ☐ describe developments in new materials and their impact

I DO — Worked example

Read how a full-mark answer is built

A child's bike helmet.

1 Identify a suitable thermoplastic for the helmet shell and explain why it is suitable. (3 marks)

Full-mark answer each ✓ is one mark

ABS ✓. It has high impact resistance, so it absorbs a knock without cracking and protects the child's head ✓. It is lightweight, so the helmet is comfortable to wear for long periods ✓.

Examiner tip: 2025 commentaries credited ABS or polypropylene with impact resistance, light weight, colours or scratch resistance.

Why marks are not awarded

"Plastic, because it is strong."

0 marks: no named thermoplastic and "strong" is not explained.

WE DO — Guided practice

Use the starters and word bank to help you

Swimming goggles have a silicone strap and seal.

2 Explain why an elastomer is suitable for the strap and seal. (2 marks)

Sentence starters

One reason is … because …

Another reason is … so …

Word bank

flexible · stretches · against · different · tightly · shapes · adjusts · comfortable

Tip: 2 marks = 2 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) Which can be reheated and reshaped?
☐ Melamine formaldehyde ☐ Urea formaldehyde ☐ Acrylic ☐ Bakelite

b) Why is nylon used for small gears?
☐ It is transparent ☐ It is self-lubricating and wear resistant ☐ It is a thermoset ☐ It is cheap foam

c) A flip-top shampoo lid with a living hinge is made from...
☐ Acrylic ☐ Polypropylene ☐ Melamine ☐ Polystyrene

d) Which plastic is used for electrical plugs?
☐ Polystyrene ☐ Urea formaldehyde ☐ Acrylic ☐ Polypropylene

4 Key terms: write the letter of the matching description next to each term.

1. **Thermoplastic**	____	**A** Sets permanently once heated; heat resistant; cannot be reshaped.
2. **Thermosetting plastic**	____	**B** Softens when heated; can be reshaped and recycled.
3. **ABS**	____	**C** Rigid, clear, polishes well. Glass substitute, signs.
4. **Acrylic**	____	**D** Thermoset electrical insulator. Plugs and sockets.
5. **Polypropylene**	____	**E** Tough, impact resistant, high gloss. Toys, casings, helmets.
6. **Urea formaldehyde**	____	**F** Fatigue resistant (living hinges), chemical resistant. Caps, crates.

5 Describe methods that could be used to identify the materials used in a product. (4 marks)

6 Give the properties and one use of Melamine formaldehyde. (2 marks)

7 Give the properties and one use of Nylon. (2 marks)

Check and reflect
Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?